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On-the-fly reduced-order modelling for skeletal reduction of gasoline surrogate combustion

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Skeletal reaction models are derived for a four-component gasoline surrogate model via a novel fully automated instantaneous local sensitivity analysis technique. This technique provides a direct approach for model reduction of large systems; in contrast to previous/ current methods constructed by ‘hands-on manipulation.’ The sensitivities of the species mass fractions and the temperature with respect to the reaction rates are estimated by a reduced-order modelling (ROM) methodology. Termed ‘implicit time-dependent basis CUR (implicit TDB-CUR),’ this methodology is based on the CUR matrix decomposition and incorporates implicit time integration to evolve the bases. The estimated sensitivities are subsequently analysed to develop skeletal reaction models with a fully automated procedure. The 1389-species gasoline surrogate model developed at Lawrence Livermore National Laboratory (LLNL) is selected as the detailed kinetics model. The skeletal reduction procedure is applied to this model in a zero-dimensional constant-pressure reactor over a wide range of initial conditions. The performances of the resulting models are evaluated by comparison against the results produced by the LLNL detailed model and also predictions by other models. Two new skeletal models consisting of 679 and 494 species are developed. The first is an alternative to an existing model with the same number of species. The predictions with this model reproduce the flame results obtained with the detailed model with less than 1% errors. The errors via the second model are less than 10%.

Keywords: Gasoline surrogate; skeletal reaction model; local sensitivity analysis; time dependent basis; CUR decomposition

1. Introduction

Gasoline fuels consist of hundreds of hydrocarbons. Computational modelling and simulation of such complex chemical mixtures require accounting for numerous species mass fractions [1]. To simplify this, gasoline surrogate mixtures are formulated with fewer hydrocarbons. These surrogates mimic the chemical kinetics and thermodynamic properties of real gasolines [2, 3]. However, surrogate models still involve $\mathcal{O}(1000)$ species, creating significant computational challenges, especially for multidimensional reacting flow simulations. Therefore, developing skeletal kinetic models for gasoline surrogates by eliminating unimportant species and reactions from the detailed models is essential for feasible simulations.

During the last two decades, a variety of techniques have been developed to reduce skeletal kinetics, by systematic elimination of unimportant species and reactions while

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maintaining the overall predictive capability [4–6]. Local sensitivity analysis (LSA), reaction flux analysis [7, 8], computational singular perturbation analysis (CSP) [9–12], and directed relation graph (DRG) and its variants [13–16] have often been used. The LSA explores the response of model outputs to small changes in parameters from their nominal values [17]. The global sensitivity analysis considers the uncertainty in the kinetic parameters (i.e. collision frequencies and activation energies) and non-linear coupling effects [18–26]. Instantaneous sensitivity analysis evaluates sensitivities at a specific time in a dynamic system, while time-independent sensitivity analysis measures how values change with parameter variations in a time-averaged sense [27, 28]. Instantaneous LSA-based techniques, which are computationally costly for large kinetics models, include methods such as the principal component analysis (PCA) [29–35], and species ranking construction [36]. The forced-optimally time dependent (f-OTD) method is specifically designed for low-rank approximations in instantaneous local sensitivity calculations. In Refs. [37–39] the f-OTD is used to approximate sensitivities to generate skeletal models for hydrocarbon combustion and nuclear burning. However, the f-OTD method is prone to ill-conditioning and requires expensive Jacobian evaluations when applied to reaction models with strong stiffness and large size. In order to enhance the computational advantages of f-OTD and to overcome its numerical challenges, a new method termed ‘implicit time-dependent basis CUR (implicit TDB-CUR)’ has been developed [40, 41]. The CUR denotes the CUR matrix decomposition, which constructs a low-rank approximation matrix in terms of a subset of its columns (C), a core matrix (U), and a subset of its rows (R) [42, 43]. The implicit TDB-CUR method combines sparse sampling, CUR decomposition, and a cost-efficient implicit time integration algorithm, offering improved efficiency and stability compared to f-OTD. These enhancements are particularly useful for dealing with stiff reaction networks characterised by reactions with orders of magnitude different time scales. Gasoline surrogate models are among these reaction networks.

The Lawrence Livermore National Laboratory (LLNL) gasoline surrogate model [44] contains the chemistry parameters of various hydrocarbon compounds including iso-octane, n-heptane, toluene and C5-C6 olefins. This model has been validated against a wide range of experimental data, making it widely applicable to predict various gasoline surrogate combustion phenomena [44–46]. However, with 1389 species and 10734 irreversible reactions, the model is computationally prohibitive for practical simulations. Mehl et al. [47] developed a 312-species model using a computer-assisted reduction mechanism, followed by a search algorithm that sequentially assessed the importance of each species. Despite its compact size, the model is not sufficiently accurate (see Section 3.2). Lu et al. [48] constructed a reasonably accurate 679-species skeletal model but still left room for further size reduction. Niemeyer et al. [49] derived two reduced models containing 97 and 79 species, each tailored to different operating conditions. Their approach incorporated a directed relation graph with error propagation and sensitivity analysis (DRGEPSA) [16], unimportant reaction elimination, isomer lumping, and analytical quasi-steady-state (QSS) assumption. Although isomer lumping and QSS significantly reduce model size, they compromise model versatility by neglecting isomer-specific reactivity differences and assuming rapid equilibration of certain species.

The objective of this work is to develop accurate skeletal models from the LLNL gasoline surrogate model using sensitivities estimated via the implicit TDB-CUR method. This reduction targets a four-component gasoline surrogate (iso-octane, n-heptane, 2-pentene, and toluene), formulated by Mehl et al. [47] and focuses on a lean-to-stoichiometric, low-temperature, homogeneous charge compression ignition (HCCI)-like range of conditions.

Table 1. Composition of the surrogate fuel.

Components	Mole fractions
iso-octane	48.70%
n-heptane	15.45%
toluene	30.60%
2-pentene	5.25%

The composition of the surrogate fuel is given in Table 1. The models are evaluated alongside two previously developed reduced models, and the detailed model to predict the ignition delay, the flame speed, and the flame structure. The reduced models are benchmarked against the detailed model rather than directly against experiments, as the detailed model has been extensively validated. The efficiency of the implicit TDB-CUR sensitivity approximation is also assessed by comparison against full-size sensitivity integration.

In summary, the main contributions of this paper are as follows: (i) a novel reduced-order modelling methodology is introduced to compute sensitivities for skeletal reduction of large, detailed models, and (ii) this methodology is applied to construct accurate skeletal kinetics models based on the LLNL gasoline surrogate model. The key feature of this approach is its automated skeletal reduction process, which highlights the generalizability of the method.

2. Methodology

The reduction procedure consists of the following: First, the implicit TDB-CUR method with the fourth-order backward differentiation formula (BDF4) [50–52] is used to compute sensitivities across multiple initial conditions (cases). The sensitivities are then collected and analysed to produce a classification of species and reactions. Finally, reduced models are constructed on the basis of these rankings.

2.1. Sensitivity analysis

As described in Refs. [37, 38, 53], local linear sensitivity coefficients in a chemical kinetics system are evaluated by infinitesimal perturbations of all reaction rates. The full-size matrix containing all the sensitivity coefficients evolves based on the following matrix differential equation:

$$\frac{dS(t)}{dt} = L(t)S(t) + F(t), \quad (1)$$

where $S(t) \in \mathbb{R}^{n_{eq} \times n_{rc}}$ is the sensitivity matrix, $L(t) \in \mathbb{R}^{n_{eq} \times n_{eq}}$ is the Jacobian matrix, and $F(t) \in \mathbb{R}^{n_{eq} \times n_{rc}}$ is the forcing term. The number of reactions is denoted by n_{rc} . The number of state variables is $n_{eq} = n_{sp} + 1$ where n_{sp} denotes the number of chemical species. The temperature is also solved as a state variable. Equation (1) represents the full-order model (FOM). The implicit TDB-CUR method aims to accurately approximate the sensitivity matrix $S^k = S(t^k)$ at all time steps. The low-rank components are stored in a singular value decomposition (SVD)-like factorised form: $S^k = U^k \Sigma^k Y^k$. To compute the low-rank components for the next time step t^{k+1} , the CUR algorithm is applied, requiring the solution of selected columns and rows of the sensitivity matrix.

In CUR decomposition, the sensitivity matrix $S(t) \in \mathbb{R}^{n_{eq} \times n_{rc}}$ is approximated by a subset of its columns and rows, $S \approx CUR$, where $C \in \mathbb{R}^{n_{eq} \times r}$ consists of r selected columns of S , $R \in \mathbb{R}^{r \times n_{rc}}$ consists of r selected rows of S , and $U \in \mathbb{R}^{r \times r}$ is a low-rank coefficient matrix that captures the interaction between the selected columns and rows. Unlike singular value decompositions, CUR provides a more interpretable decomposition, as the selected columns and rows correspond directly to actual sensitivity information rather than abstract linear combinations. In the implicit TDB-CUR, this decomposition is applied to the sensitivity matrix at each time step, maintaining a low-rank representation without requiring additional evolving equations for the low-rank components. The columns and rows are integrated implicitly using the BDF4 method. The updated sensitivities are stored in the same format: $S^{k+1} = U^{k+1} \Sigma^{k+1} Y^{k+1}$. To obtain the initial conditions of the low-rank components, the FOM is integrated for several time steps. Then the full-order sensitivity matrix $S^o = S(t^o)$ is factorised by an SVD. This factorisation leads to three low-rank components: $U^o \in \mathbb{R}^{n_{eq} \times r}$, $\Sigma^o \in \mathbb{R}^{r \times r}$, and $Y^o \in \mathbb{R}^{n_{rc} \times r}$, such that $S^o \approx U^o \Sigma^o Y^{oT}$, where r denotes the number of retained modes ($r \ll n_{eq}$). Then, instead of integrating the FOM, the three low-rank components are updated and saved at each time step. In this way, the computing and memory cost of implicit sensitivities integration is substantially reduced in comparison to that required for the solution of the FOM.

For comparison, the f-OTD formulation introduces additional evolution equations that involve inversion of a correlation matrix, which becomes ill-conditioned in the presence of small singular values. Moreover, implicit integration of the f-OTD system requires the evaluation of a Jacobian of size $(n_{eq} + n_{rc}) \times (n_{eq} + n_{rc})$ at every time step (12, 124 $\times$ 12, 124 in the gasoline case), making the computations prohibitively expensive.

In TDB-CUR, the integration of selected columns and rows involves solving the following equations:

$$S^{k+1}(:, I_c) = S^k(:, I_c) + \delta S^k(:, I_c), \quad (2a)$$

$$S^{k+1}(I_r, :) = S^k(I_r, :) + \delta S^k(I_r, :), \quad (2b)$$

where $\delta S^k(:, I_c)$ and $\delta S^k(I_r, :)$ denote the correction terms for the columns I_c and rows I_r , respectively. Matlab-style indexing is adopted, where $S^k(:, I_c)$ denotes a submatrix containing the specified columns, and $S^k(I_r, :)$ denotes a submatrix containing the specified rows. The GappyPOD + E method [54] is used to select I_c and I_r to minimise residuals from the low-rank approximation. The column correction term is obtained by solving the linear system:

$$A^k \delta S^k(:, I_c) = b^k, \quad (3)$$

where A^k and b^k depend on the implicit integrator. For the BDF4 scheme: $A^k = \left(I - \frac{12}{25} \Delta t L \right)$ and $b^k = \left[\frac{48}{25} S^k(:, I_c) - \frac{36}{25} S^{k-1}(:, I_c) + \frac{16}{25} S^{k-2}(:, I_c) - \frac{3}{25} S^{k-3}(:, I_c) \right] - S_i^{k+1}(:, I_c) + \frac{12}{25} \Delta t (L^k S^k(:, I_c) + F^k(:, I_c))$, where I denotes the identity matrix. Since the local sensitivity system is linear, only one linear solve is required. For the FOM, the same formula is used, except that I_c includes all columns.

Row integration requires special treatment. In a linear system of equations, the solution for any entry in the unknown vector is coupled to all other entries in that vector. Therefore, integrating the selected rows requires solving the entire linear system. To reduce computational cost, a low-rank approximation of $\delta S^k(I_r, :)$ is used in implicit TDB-CUR. Resulting

a reduced-order linear system:

$$A_{(r)}^k \delta S^k(I_r, :) = b_{(r)}^k, \quad (4)$$

where $A_{(r)}^k = A^k(I_r, I_r) + A^k(I_r, \tilde{I}_r)U_\sigma^k(\tilde{I}_r, :)U_\sigma^k(I_r, :)^\dagger$ and $b_{(r)}^k = \left[\frac{48}{25}S^k(I_r, :) - \frac{36}{25}S^{k-1}(I_r, :) + \frac{16}{25}S^{k-2}(I_r, :) - \frac{3}{25}S^{k-3}(I_r, :) \right] - S_i^{k+1}(I_r, :) + \frac{12}{25}\Delta t(L^k(I_r, :)S^k + F^k(I_r, :))$. $\tilde{I}_r$ denotes the complement set of I_r and the † sign denotes a pseudo-inverse. The orthonormal basis U_σ^k is obtained by a SVD operation on a matrix S_σ^k that is obtained by subtracting previous column solutions from the t^{k+1} column solution: $S_\sigma^k = S^{k+1}(:, I_c) - S^k(:, I_c) - S^{k-1}(:, I_c) - S^{k-2}(:, I_c) - S^{k-3}(:, I_c)$. Equation (4), with $A_{(r)}^k \in \mathbb{R}^{r \times r}$, can be solved efficiently to find $\delta S^k(I_r, :)$.

The low-rank components of the sensitivity matrix at t^{k+1} are computed as follows:

$$S^{k+1}(:, I_c) = U \Sigma Y^T, \quad (5a)$$

$$V = [U(I_r, :)]^\dagger S^{k+1}(I_r, :)]^T, \quad (5b)$$

$$V = Y^{k+1} \Sigma^{k+1} R_U, \quad (5c)$$

$$U^{k+1} = U R_U, \quad (5d)$$

where U is computed via an SVD on column solutions. Σ^{k+1} and Y^{k+1} are obtained by SVD in matrix V . Finally, the remaining sensitivity component U^{k+1} is obtained by rotating the basis U .

2.2. Reactions and species ranking

After collecting low-rank sensitivity components from multiple cases, the reactions and the species can be ranked. The right singular vectors $Y^k = Y(t^k)$, which reflect the importance of the reaction, are weighted by the singular values Σ^k :

$$W_{(j)}^k = \sum_n |Y_n^k| \Sigma_n^k / \sum_n \Sigma_n^k, \quad (6)$$

where the index n denotes the n th right singular vector or singular value. Each element of $W_{(j)}^k \in \mathbb{R}^{n_r}$, is positive and is associated with a certain reaction. The higher the associated W value, the more important the reaction is. The parameter $\chi \in \mathbb{R}^{n_r}$ is defined as the highest value associated with W across all resolved time-steps and cases. The elements of the χ vector are sorted in descending order to find the indices of the most important reactions in the detailed model. The species are ranked according to their first presence in the sorted reactions, i.e. a species that first appears in a higher-ranked reaction is considered more important than one that first participates in a lower-ranked reaction. This process results in a reaction and a species ranking based on the χ vector. The reduced model is constructed by removing the less important species from the species hierarchy while retaining all reactions associated with the remaining species.

The skeletal reduction scheme shown here is based on instantaneous local sensitivities. Unlike PCA-based sensitivity methods [30], where full sensitivities are computed first and then reduced, the implicit TDB-CUR approach solves for the sensitivities directly in their reduced form. This leads to computational savings in floating-point operations (FLOPs) and substantial reductions in memory usage, while also eliminating the need to store full-size sensitivity data.

3. Skeletal reduction of gasoline surrogate

The implicit TDB-CUR method with the BDF4 integrator [52] is employed for low-rank approximation of gasoline surrogate model sensitivities. A total of 11 cases are considered, with the initial conditions summarised in Table 2 taken from Refs. [47, 49]. These cases cover the ranges of 750 K to 1200 K for the temperature, 10 atm to 60 atm for the pressure, and 0.2 to 1.0 for the equivalence ratio. The CUR rank is set as $r = 7$. To benchmark the efficiency and accuracy of implicit TDB-CUR, the FOM is also solved for all cases by the BDF4. Following the FOM solution, an SVD of the sensitivity matrix is performed at each time step. The data from these SVD are referred to as the instantaneous SVD (iSVD). The seven largest singular values from the iSVD are compared with those obtained using the implicit TDB-CUR. The sensitivities estimated by the implicit TDB-CUR are further analysed to generate species and reaction rankings. The performance of the reduced models

Table 2. Clock times for computing sensitivities.

	Initial conditions	Time steps	FOM	Implicit TDB-CUR
Case 1	T800, P10, ϕ 1.0	6466	6320.6 s	3587.3 s
Case 2	T750, P60, ϕ 1.0	6826	6905.2 s	3874.6 s
Case 3	T1200, P60, ϕ 0.6	4544	3633.8 s	2080.0 s
Case 4	T1100, P10, ϕ 0.6	4766	3891.5 s	2210.6 s
Case 5	T1000, P60, ϕ 0.6	4948	4538.6 s	2575.6 s
Case 6	T800, P10, ϕ 0.6	6566	6239.1 s	3593.2 s
Case 7	T750, P10, ϕ 0.6	6820	6629.3 s	3842.9 s
Case 8	T750, P60, ϕ 0.6	6764	6774.1 s	3847.3 s
Case 9	T800, P60, ϕ 0.2	4440	3031.3 s	1745.5 s
Case 10	T700, P20, ϕ 0.2	4852	3209.0 s	1838.3 s
Case 11	T800, P20, ϕ 0.2	4244	3435.2 s	1956.3 s

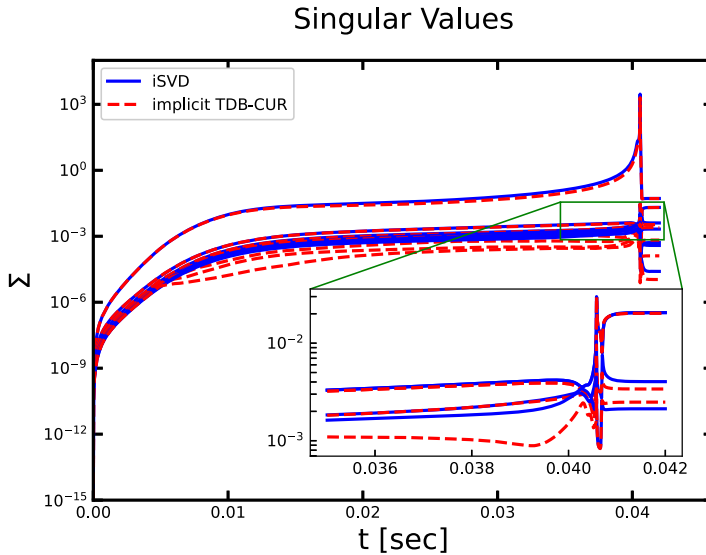


Figure 1. Singular values in case 1 ($\phi = 1.0$, $T = 800$ K, $P = 10$ atm).

is evaluated by comparison against the results of the detailed model. Two existing reduced models are also included for comparison: *chem312* [47] and *chem679* [48].

3.1. TDB-CUR evaluation

Figure 1 provides a comparison of the singular values of the sensitivity matrix obtained by implicit TDB-CUR against those via iSVD for Case 1. It is shown that there is one

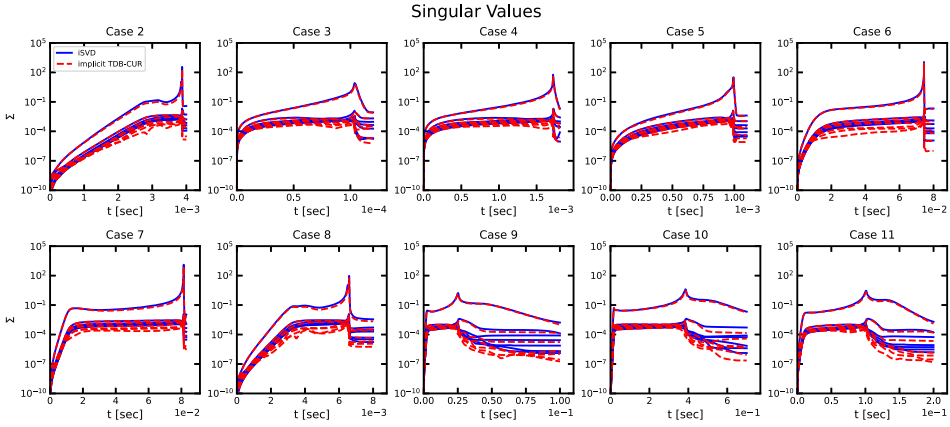


Figure 2. Singular values in cases 2–11.

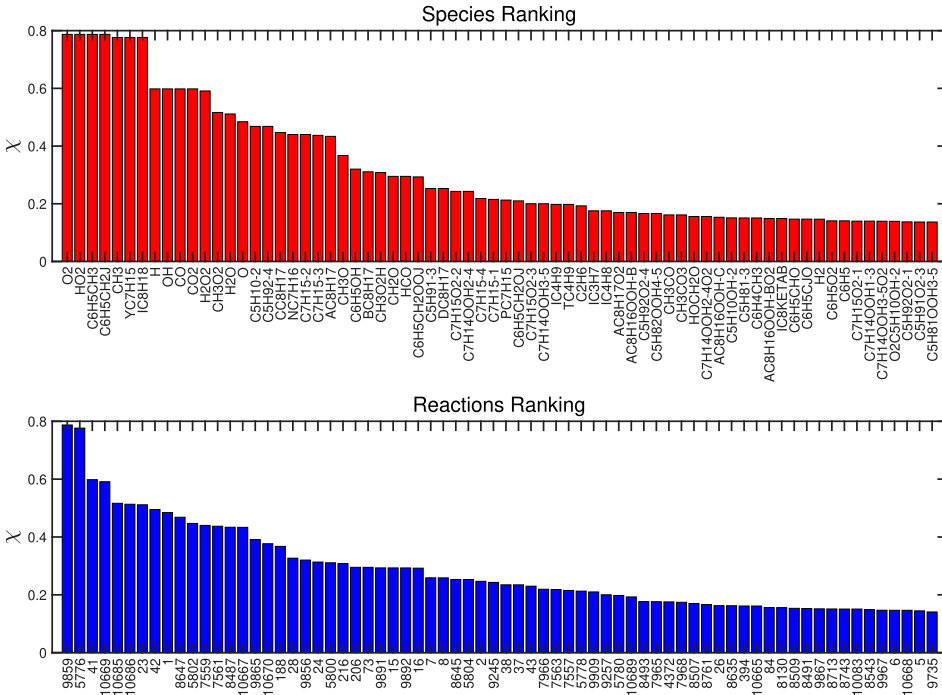


Figure 3. Importance ranking of the species and reactions. Reaction ranking is labelled by the reaction indices associated with the supplemental Cantera file.

dominant mode for which the singular value is multiple orders of magnitude larger than the others. The implicit TDB-CUR is able to accurately capture the first three singular values. The smaller modes do not align with the FOM results, but have no significant effects on the predicted results, since the right singular values are weighted by the singular values. Figure 2 presents results for Cases 2–11, showing that the leading singular values are consistently captured with high accuracy across all conditions. As for the efficiency of

Table 3. Chemistry models.

Model	n_{sp}	n_{rc}
Detailed	1389	10734
rd679	679	5731
rd494	494	4154

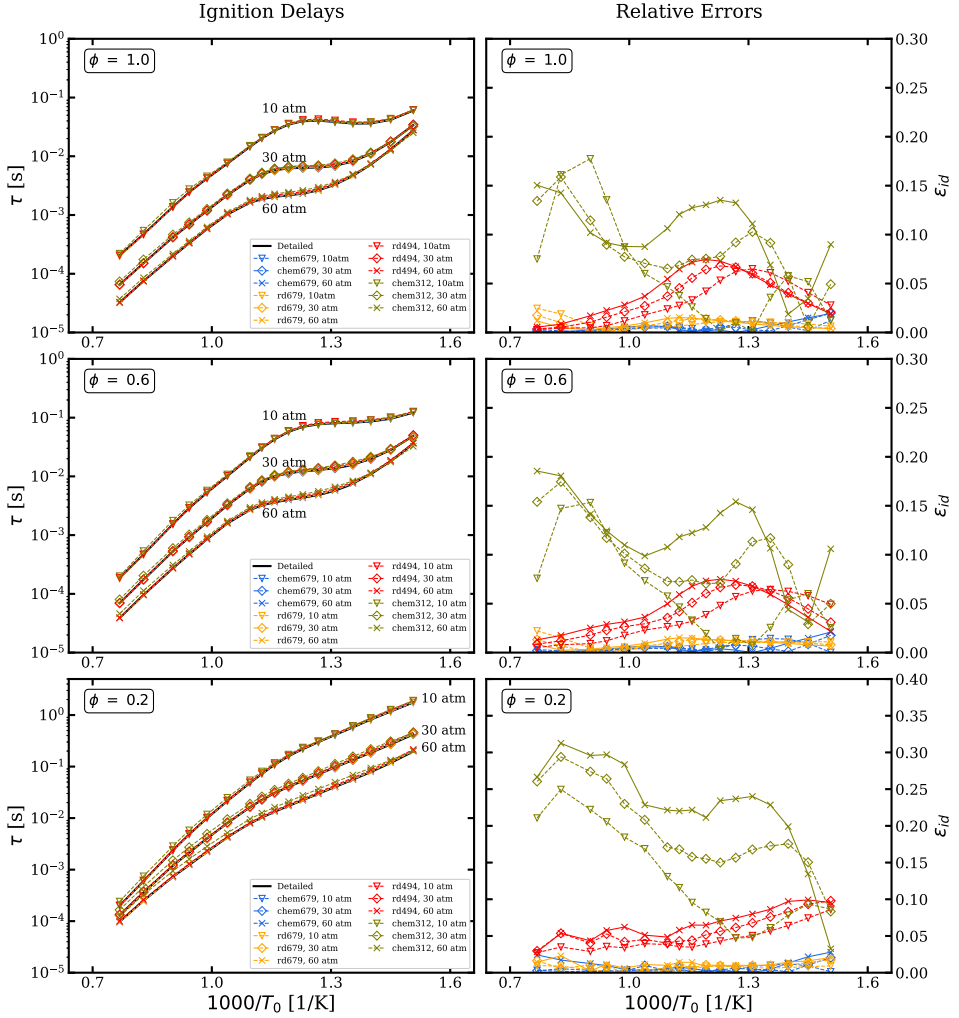


Figure 4. Left: Ignition delay test of the reduced models. Right: Relative errors $\epsilon_{id} = |\tau_{reduced} - \tau_{detailed}| / \tau_{detailed}$.

the implicit TDB-CUR, the clock time of ROM simulation is compared against FOM in Table 2. In all the cases, the ROM requires less than 60% time of the FOM.

3.2. Skeletal model validation

In Figure 3, the top 70 most important species and reactions of the LLNL gasoline surrogate model are ranked based on their importance. The higher the χ values, the greater the importance of a species/reaction. The Appendix provide the complete ranking of the species and rankings. In the reactions ranking, a toluene oxidation reaction (*Reaction 9859*: $C_6H_5CH_3 + O_2 \rightarrow C_6H_5CH_2J + HO_2$) is identified as the most important. The species involved in this reaction are identified as the most important species, as indicated by the species ranking. The second most important is an iso-octane pyrolysis reaction (*Reaction 5776*: $IC_8H_{18} \rightarrow CH_3 + YC_7H_{15}$). This is consistent with the findings in Ref. [55]. Based on the species ranking, two reduced reaction models are suggested. The first contains the top 679 most important species (*rd679*), the same number as in *chem679* [48]. This model yields less than %1 error. The second model with the 494 species that dominate (*rd494*) guarantees 10% maximum relative error of all comparative assessments with the minimum number of species. The numbers of reactions for these reduced models are listed in Table 3. The reaction size refers to the number of irreversible reactions.

In Figure 4, the ignition delays are shown for several equivalence ratios and pressure values. The relative errors in predicting the delays are also provided and indicate that *rd679* and *chem679* provide the best overall accuracy. The relative errors associated with these two models are of the order 1%. For the *rd494* model, the relative errors are less than 8% in most cases. With $\phi = 0.6$ & 1.0, the *rd494* models show larger errors when $1000/T_0 = 1.1 \sim 1.4 \text{ K}^{-1}$, which correspond to regions with lower gradients. This behaviour is known as the ‘negative temperature coefficients’ (NTC) [56]. The shape of the *rd494* relative error curves indicates that the NTC can increase the difficulty of producing accurate reduced models using sensitivity coefficients. The *chem312* model [47] provides errors up to 30%, but produces accurate predictions when $P = 10 \text{ atm}$ and NTC behaviour occurs.

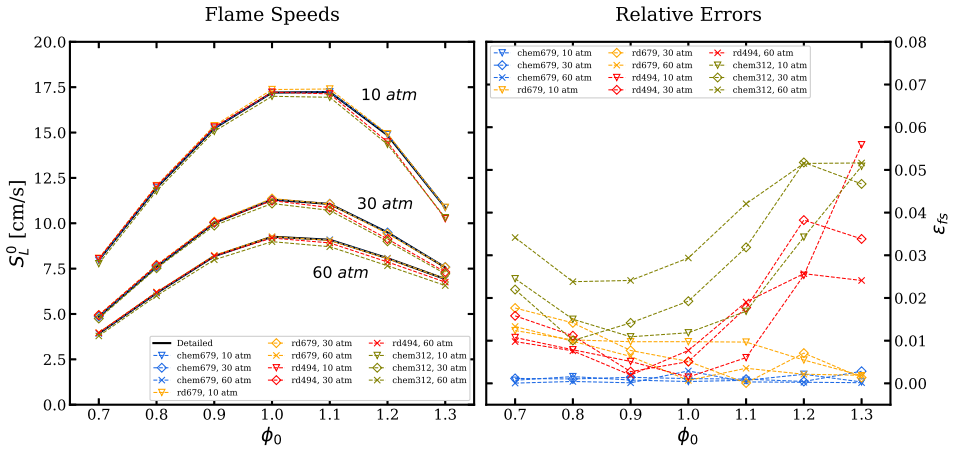


Figure 5. Left: Laminar flame speed test ($T_0 = 300 \text{ K}$). Right: Relative errors $\epsilon_{fs} = |S_{L, \text{reduced}}^0 - S_{L, \text{detailed}}^0| / S_{L, \text{detailed}}^0$.

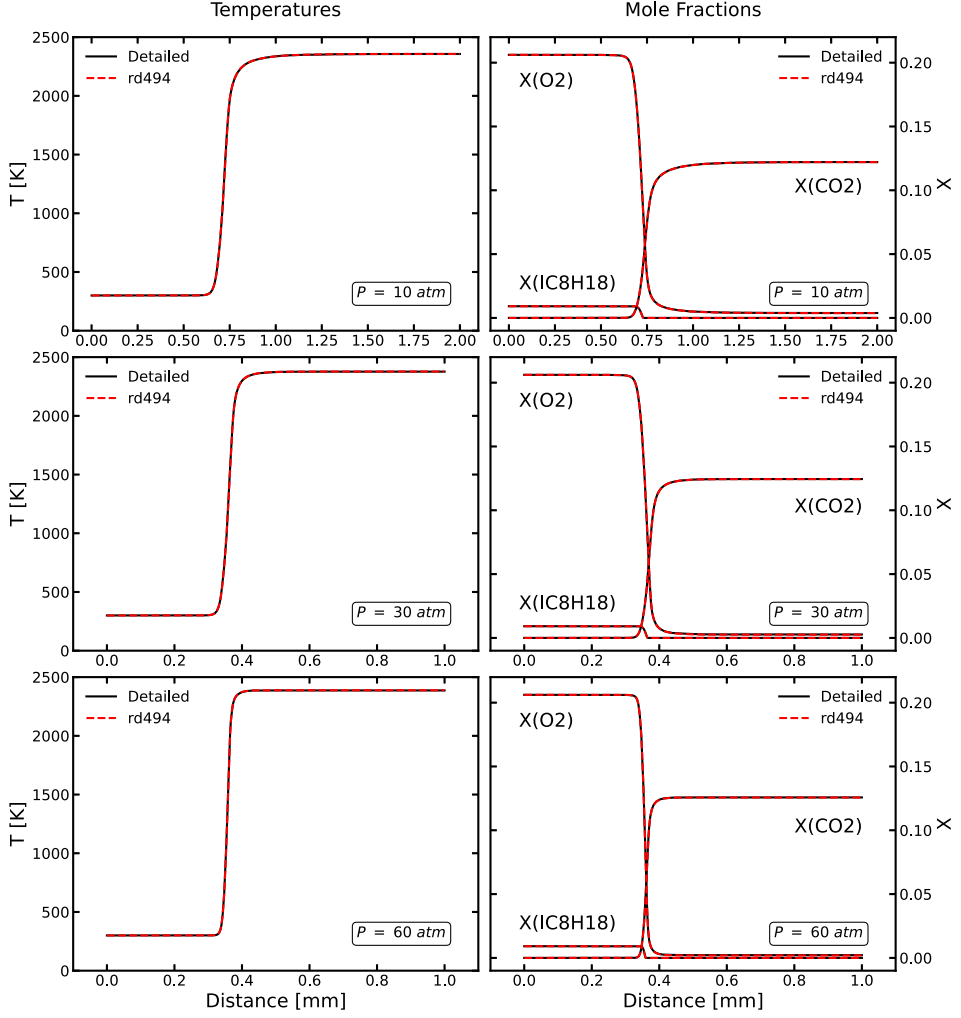


Figure 6. 1D flame structure test of the *rd494* model. Left: The distribution of flame temperatures. Right: Mole fractions of key species.

Simulations are also conducted to determine the laminar flame speed with $T_0 = 300\text{ K}$. Figure 5 shows that all relative errors are within 6%. The *rd494* model provides less than the 3% error in most cases, while *rd679* yields less than 2% error for all cases. The 1D flame structures are determined for several pressure values. In Figure 6, the distribution of temperatures and mole fractions of O_2 , CO_2 , and IC_8H_{18} corroborate that the *rd494* model perfectly replicates the 1D flame structure.

4. Conclusions

A new methodology is developed for reduced-order skeletal modelling of gasoline surrogate combustion. The novelty of the method is its fully automated mechanism which portrays a significant edge over existing ‘hands-on methods’. The implicit time-dependent

basis CUR (implicit TDB-CUR) method is employed for low-rank approximation of linear sensitivities in the detailed combustion model of the Lawrence Livermore National Laboratory (LLNL). It is demonstrated that this new method evaluates sensitivities with computational costs significantly lower than that required to solve the full-order model (FOM). The computed sensitivities are then analysed for skeletal reduction of the LLNL model. In doing so, the species are ranked based on their sensitivities to the reaction rates. The *rd679* skeletal model developed here shows the same precision as the *chem679* model developed in Ref. [48]. The *rd494* with a lower number of species is also generated. The accuracy and efficiency of the model are assessed by comparing its predictions for the ignition delay, the flame speed, and 1-D flame structure. The *rd494* model, which contains about 35% of the total species suggested by the LLNL model, while yielding better than 90% accuracy, is a reliable and affordable alternative to the detailed model. Although the current models are larger than typically required for engine-scale simulations, they serve as a foundation for future refinements for future practical applications. Further reduction in model size can be achieved by incorporating additional assumptions, such as isomer lumping or quasi-steady-state approximations, depending on the accuracy and efficiency requirements in specific applications. The relatively larger errors observed for the *rd494* model in the NTC region highlight the importance of low-temperature oxidation pathways. Future work leveraging sensitivity information for pathway-level analyses is recommended to provide more detailed insight into the underlying kinetics of NTC behaviour.

It is demonstrated that the implicit TDB-CUR method accurately calculates the sensitivities and generates skeletal reaction models in an automated and efficient manner. The ability to perform such a reduction automatically for large and complex systems eliminates the need for user expertise in chemical kinetics networks, marking a significant advancement. Although the reduction scheme is applied to a specific gasoline surrogate mixture, it is broadly applicable to the reduction of other multi-component fuels. Moreover, it is well suited to reduce other reaction networks, such as nuclear [57] and biochemical systems [58].

Disclosure statement

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Appendix. Reduced models

All reduced models can be accessed through this [link](#).